

Your Code

```
1 n=int(input())
2 if n<=1:
3     print("Not a prime number")
4 else:
5     prime=True
6     for i in range(2, int(n**0.5)+1):
7         if n%i==0:
8             prime=False
9             break
10    if prime:
11        print("prime number")
12    else:
13        print("not a prime number")
```

Input

7

Output

prime number

Q2 Fibonacci series

10 marks

A numerical computation module is designed to generate a sequence where each number is the sum of the two preceding numbers. This sequence is known as the Fibonacci series.

Write a Python program that:

- Accepts a number n from the user
- Generates and displays the first n terms of the Fibonacci sequence
- Uses looping constructs to compute the sequence

Ensure the program handles cases where n is less than or equal to zero.

Your Code

```
1 n=int(input())
2 if n<=0:
3     print("Invalid input")
4 else:
5     a=0
6     b=1
7     for i in range(n):
8         print(a, end=" ")
9         c=a+b
10        a=b
11        b=c
```

Input

5

Output

0 1 1 2 3

Your Code

```
1 s=input()
2 v=c=d=sp=0
3 for i in s:
4     if i in "aeiouAEIOU":
5         v+=1
6     elif i.isalpha():
7         c+=1
8     elif i.isdigit():
9         d+=1
10    else:
11        sp+=1
12 print(v)
13 print(c)
14 print(d)
15 print(sp)
```

Input

Hello123@#

Output

2
3
3
2

Your Code

```
1 num=int(input())
2 if num%2==0:
3     print("even")
4 else:
5     print("odd")
```

Input

2

Output

even

Q5 Print numbers from 1 to 10

10 marks

A program is intended to display numbers from 1 to 10 using a loop. However, some parts of the code are missing. Complete the program by filling in the blanks so that it correctly prints numbers from 1 to 10.

Program to print numbers from 1 to 10

for i in _____:

print(_____)

Also explain how the range() function works in this context.

Your Code

```
1 for i in range(1,11):
2     print(i)
```

Input

stdin input

Output

1
2
3
4
5
6